

Worked examples for Lecture 16: Ray theory

David Al-Attar

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These examples accompany Lecture 16. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: The isotropic eikonal equation and a linear velocity gradient. (a) Show that in an isotropic medium with P-wave speed $\alpha(\mathbf{x})$ the eikonal equation for the P-wave sheet of the local slowness surface reduces to $\|\nabla T\| = 1/\alpha$, and that the corresponding ray Hamiltonian is $H(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\alpha(\mathbf{x})^2\|\mathbf{p}\|^2$. (b) Consider a two-dimensional model with horizontal coordinate x , depth z measured positively downwards, and P-wave speed $\alpha(z) = \alpha_0 + gz$ with $\alpha_0 > 0$ and $g > 0$. A ray leaves the origin with take-off angle θ_0 measured from the downward vertical. Solve Hamilton's equations for this ray. Show that the horizontal slowness $p_x = \sin \theta_0 / \alpha_0$ is conserved, that the ray is a circular arc of radius $R = 1/(gp_x)$ centred at the depth $z = -\alpha_0/g$, and that the travel time along the ray is

$$T = \frac{1}{g} \ln \left[\frac{\tan(\theta/2)}{\tan(\theta_0/2)} \right],$$

where θ is the angle the ray makes with the downward vertical at the point in question.

(c) Find the horizontal distance X at which the ray returns to the surface $z = 0$ and the travel time $T(X)$, and evaluate both for $\alpha_0 = 5 \text{ km s}^{-1}$, $g = 0.05 \text{ s}^{-1}$ and $\theta_0 = 30^\circ$.

Solution. (a) In an isotropic medium the elastic tensor is $A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$, and so the local Christoffel operator defined in the lecture takes the form

$$\Gamma_{ik}(\mathbf{x}, \mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l = \alpha^2 p_i p_k + \beta^2 (\|\mathbf{p}\|^2 \delta_{ik} - p_i p_k), \quad (1)$$

with $\alpha^2 = (\lambda + 2\mu)/\rho$ and $\beta^2 = \mu/\rho$, exactly as for plane waves in Lecture 15 but with ρ , λ and μ now functions of $\mathbf{x}$. The two terms are, respectively, $\|\mathbf{p}\|^2$ times the projections parallel and orthogonal to $\hat{\mathbf{p}}$, so the eigenvalues of Γ are $\alpha^2\|\mathbf{p}\|^2$, with eigenvector $\hat{\mathbf{p}}$, and $\beta^2\|\mathbf{p}\|^2$, with the two-dimensional eigenspace orthogonal to $\hat{\mathbf{p}}$. The eikonal equation $\det(\Gamma - \mathbf{I}) = 0$ therefore factorises as

$$(\alpha^2\|\mathbf{p}\|^2 - 1) (\beta^2\|\mathbf{p}\|^2 - 1)^2 = 0, \quad (2)$$

which is the lecture's general factorised form with the local phase speeds $c_1 = \alpha$ and $c_2 = c_3 = \beta$ independent of the direction $\hat{\mathbf{p}}$. Selecting the P-wave sheet and recalling

that $\mathbf{p} = \nabla T$, the reduced eikonal equation is

$$\alpha(\mathbf{x})^2 \|\nabla T(\mathbf{x})\|^2 = 1, \quad \text{i.e.} \quad \|\nabla T\| = \frac{1}{\alpha}, \quad (3)$$

and the ray Hamiltonian $H = \frac{1}{2} \|\mathbf{p}\|^2 c_1^2$ of the lecture becomes

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2} \alpha(\mathbf{x})^2 \|\mathbf{p}\|^2, \quad (4)$$

with the eikonal equation being $H[\mathbf{x}, \nabla T(\mathbf{x})] = \frac{1}{2}$. The polarisation $\hat{\mathbf{a}}^0 = \hat{\mathbf{p}}$ is, as for plane waves, along the local direction of propagation.

(b) We work in the (x, z) -plane, with $\mathbf{x} = (x, z)$ and $\mathbf{p} = (p_x, p_z)$, and the sign conventions stated in the question: z increases downwards, so that α increases with depth, and a ray travelling straight down has $\theta = 0$. The initial point in phase space is $\mathbf{x}^0 = (0, 0)$ together with

$$\mathbf{p}^0 = \frac{1}{\alpha_0} (\sin \theta_0, \cos \theta_0), \quad (5)$$

which has $\|\mathbf{p}^0\| = 1/\alpha_0$ and so satisfies $H = \frac{1}{2}$ as required. From eq. (4), $\partial H / \partial p_i = \alpha^2 p_i$ and $\partial H / \partial x_i = \alpha \|\mathbf{p}\|^2 \partial \alpha / \partial x_i$, and since α depends only on z with $d\alpha/dz = g$, Hamilton's canonical equations read

$$\frac{dx}{d\sigma} = \alpha^2 p_x, \quad \frac{dz}{d\sigma} = \alpha^2 p_z, \quad \frac{dp_x}{d\sigma} = 0, \quad \frac{dp_z}{d\sigma} = -\|\mathbf{p}\|^2 \alpha g = -\frac{g}{\alpha}, \quad (6)$$

where in the last equation we have used the conservation of the Hamiltonian, $\alpha^2 \|\mathbf{p}\|^2 = 1$, along the solution. The third equation shows at once that the horizontal slowness is conserved,

$$p_x = \frac{\sin \theta_0}{\alpha_0} \equiv p, \quad (7)$$

which is the conserved quantity associated with the translational symmetry of the model in x ; it is often called the ray parameter. Because $\|\mathbf{p}\| = 1/\alpha$ everywhere along the ray, we can introduce an angle $\theta(\sigma)$ through

$$p_x = \frac{\sin \theta}{\alpha}, \quad p_z = \frac{\cos \theta}{\alpha}, \quad (8)$$

with $\theta(0) = \theta_0$. The first two equations in eq. (6) then give $d\mathbf{x}/d\sigma = \alpha(\sin \theta, \cos \theta)$, so θ is precisely the angle between the ray and the downward vertical, and $ds/d\sigma = \alpha$ for the arc length s along the ray. Combining eqs. (7) and (8) we obtain Snell's law for this model,

$$\frac{\sin \theta}{\alpha(z)} = p, \quad (9)$$

so that θ increases with depth: the ray turns away from the vertical as it enters faster material. Differentiating eq. (9) with respect to σ and using $dz/d\sigma = \alpha \cos \theta$ gives $\cos \theta d\theta/d\sigma = pg\alpha \cos \theta$, and hence

$$\frac{d\theta}{d\sigma} = pg\alpha = g \sin \theta. \quad (10)$$

It is now simplest to use θ rather than σ as the parameter along the ray. Dividing $dx/d\sigma = \alpha \sin \theta$ and $dz/d\sigma = \alpha \cos \theta$ by eq. (10), and writing $R = 1/(gp)$, we find

$$\frac{dx}{d\theta} = \frac{\alpha}{g} = R \sin \theta, \quad \frac{dz}{d\theta} = \frac{\alpha \cos \theta}{g \sin \theta} = R \cos \theta, \quad (11)$$

where we have used $\alpha = \sin \theta / p$ from eq. (9). Integrating from θ_0 , where the ray is at the origin,

$$x = R(\cos \theta_0 - \cos \theta), \quad z = R(\sin \theta - \sin \theta_0), \quad (12)$$

and therefore

$$(x - R \cos \theta_0)^2 + (z + R \sin \theta_0)^2 = R^2. \quad (13)$$

The ray is a circle of radius

$$R = \frac{1}{gp} = \frac{\alpha_0}{g \sin \theta_0}, \quad (14)$$

whose centre lies at horizontal position $R \cos \theta_0$ and at the depth $-R \sin \theta_0 = -\alpha_0/g$. This depth is the same for all take-off angles: it is the height above the surface at which the linear velocity profile would extrapolate to zero. Note also that eq. (11) gives $ds/d\theta = R$, so the curvature $d\theta/ds = 1/R$ is indeed constant along the ray.

For the travel time we use the lecture's result that the generating parameter is the travel time, $T[\mathbf{x}(\sigma)] = \sigma$, together with eq. (10):

$$T = \int_0^\sigma d\sigma' = \int_{\theta_0}^\theta \frac{d\theta'}{g \sin \theta'} = \frac{1}{g} \ln \left[\frac{\tan(\theta/2)}{\tan(\theta_0/2)} \right], \quad (15)$$

where we have used $\frac{d}{d\theta} \ln \tan(\theta/2) = 1/\sin \theta$. The same result follows from $dT = ds/\alpha = R d\theta/\alpha$ and eq. (9).

(c) The ray is horizontal, $\theta = \pi/2$, at its turning point, which from eq. (12) lies at the depth

$$z_t = R(1 - \sin \theta_0) = \frac{\alpha_0}{g} \left(\frac{1}{\sin \theta_0} - 1 \right), \quad (16)$$

and by the symmetry of the circle it returns to $z = 0$ when $\theta = \pi - \theta_0$, at the distance

$$X = 2R \cos \theta_0 = \frac{2\alpha_0}{g} \cot \theta_0. \quad (17)$$

Setting $\theta = \pi - \theta_0$ in eq. (15) and using $\tan[(\pi - \theta_0)/2] = \cot(\theta_0/2)$,

$$T(X) = \frac{2}{g} \ln \cot \frac{\theta_0}{2} = \frac{2}{g} \operatorname{arcsinh} \left(\frac{gX}{2\alpha_0} \right), \quad (18)$$

where the second form follows from $\cot(\theta_0/2) = \cot \theta_0 + \sqrt{1 + \cot^2 \theta_0}$ and $\operatorname{arcsinh} u = \ln(u + \sqrt{1 + u^2})$, with $u = \cot \theta_0 = gX/(2\alpha_0)$ from eq. (17). Eq. (18) is the travel-time curve for this model. For small X it reduces to $T \approx X/\alpha_0$, while for large X the travel time grows only logarithmically because the ray spends most of its time in fast material at depth. For the given numbers, $p = 0.1 \text{ s km}^{-1}$, $R = 200 \text{ km}$, the centre of the circle lies at $(173.2, -100) \text{ km}$, the turning depth is $z_t = 100 \text{ km}$, the ray returns to the surface at $X = 346.4 \text{ km}$, and $T(X) = 40 \ln \cot 15^\circ = 52.68 \text{ s}$. This is considerably less than the 69.28 s that would be taken along the straight line at speed α_0 . Integrating

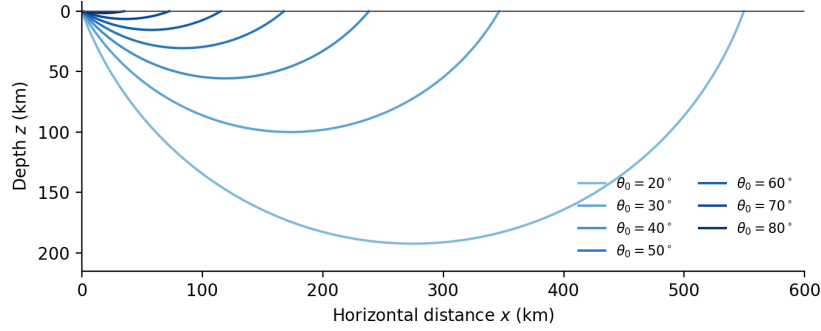


Fig. 1: Fan of rays leaving the origin in the model $\alpha(z) = \alpha_0 + gz$ with $\alpha_0 = 5 \text{ km s}^{-1}$ and $g = 0.05 \text{ s}^{-1}$, for take-off angles between 20° and 80° . Each ray is a circular arc centred at the depth $-\alpha_0/g = -100 \text{ km}$.

Hamilton's equations in eq. (6) numerically with `scipy.integrate.solve_ivp` reproduces the circle to 10^{-7} km and the travel time to 10^{-8} s . Fig. 1 shows the resulting fan of rays.

This example is the simplest illustration of how a velocity that increases with depth returns rays to the surface, which is the mechanism that makes body-wave seismology possible; the spherical analogue is developed in Lecture 17.

Example 2: From Hamilton's equations to the ray equation. For the isotropic P-wave Hamiltonian $H(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\alpha(\mathbf{x})^2\|\mathbf{p}\|^2$ write down Hamilton's equations. Show that the arc length s along a ray satisfies $ds/d\sigma = \alpha$, and hence that rays satisfy the ray equation

$$\frac{d}{ds} \left(\frac{1}{\alpha} \frac{d\mathbf{x}}{ds} \right) = \nabla \left(\frac{1}{\alpha} \right).$$

Deduce that rays are straight lines when α is constant, and show that in general a ray has curvature $\kappa = \|\hat{\mathbf{t}} \times \nabla \ln \alpha\|$, where $\hat{\mathbf{t}}$ is the unit tangent, and bends towards regions of lower wave speed. Check the result against Example 1.

Solution. The partial derivatives of the Hamiltonian are $\partial H / \partial p_i = \alpha^2 p_i$ and $\partial H / \partial x_i = \alpha \|\mathbf{p}\|^2 \partial \alpha / \partial x_i$, and so Hamilton's equations are

$$\frac{d\mathbf{x}}{d\sigma} = \alpha^2 \mathbf{p}, \quad \frac{d\mathbf{p}}{d\sigma} = -\|\mathbf{p}\|^2 \alpha \nabla \alpha. \quad (19)$$

Along a solution the Hamiltonian is conserved and equal to $\frac{1}{2}$, so $\|\mathbf{p}\| = 1/\alpha$, and the equations simplify to

$$\frac{d\mathbf{x}}{d\sigma} = \alpha \hat{\mathbf{p}}, \quad \frac{d\mathbf{p}}{d\sigma} = -\frac{\nabla \alpha}{\alpha} = -\nabla \ln \alpha. \quad (20)$$

The first of these shows that the ray is everywhere tangent to the local slowness vector, and that

$$\frac{ds}{d\sigma} = \left\| \frac{d\mathbf{x}}{d\sigma} \right\| = \alpha, \quad (21)$$

which is just the statement $dT = ds/\alpha$ that the travel time accumulates at the local slowness. The unit tangent is therefore

$$\hat{\mathbf{t}} = \frac{d\mathbf{x}}{ds} = \frac{1}{\alpha} \frac{d\mathbf{x}}{d\sigma} = \hat{\mathbf{p}}, \quad \text{so that} \quad \mathbf{p} = \frac{1}{\alpha} \frac{d\mathbf{x}}{ds}. \quad (22)$$

Differentiating this expression with respect to s and using eqs. (20) and (21),

$$\frac{d}{ds} \left(\frac{1}{\alpha} \frac{d\mathbf{x}}{ds} \right) = \frac{d\mathbf{p}}{ds} = \frac{1}{\alpha} \frac{d\mathbf{p}}{d\sigma} = -\frac{\nabla\alpha}{\alpha^2} = \nabla \left(\frac{1}{\alpha} \right), \quad (23)$$

which is the ray equation. It is a second-order equation for the ray path alone, with the slowness vector eliminated; it is also the Euler–Lagrange equation for Fermat’s principle of stationary travel time $\int ds/\alpha$, though we do not need that here. If α is constant the right hand side vanishes and $d^2\mathbf{x}/ds^2 = \mathbf{0}$, so rays are straight lines, as they must be if the theory is to reproduce plane waves in a homogeneous body.

To find the curvature we expand the left hand side of eq. (23), using the chain rule in the form $d(1/\alpha)/ds = \hat{\mathbf{t}} \cdot \nabla(1/\alpha)$, which gives

$$\frac{1}{\alpha} \frac{d\hat{\mathbf{t}}}{ds} + \hat{\mathbf{t}} \left[\hat{\mathbf{t}} \cdot \nabla \left(\frac{1}{\alpha} \right) \right] = \nabla \left(\frac{1}{\alpha} \right). \quad (24)$$

Multiplying by α and using $\alpha \nabla(1/\alpha) = -\nabla \ln \alpha$,

$$\frac{d\hat{\mathbf{t}}}{ds} = - \left[\nabla \ln \alpha - \hat{\mathbf{t}}(\hat{\mathbf{t}} \cdot \nabla \ln \alpha) \right] = -(\nabla \ln \alpha)_{\perp}, \quad (25)$$

where $(\nabla \ln \alpha)_{\perp}$ denotes the component of $\nabla \ln \alpha$ orthogonal to the ray. By the Frenet formula $d\hat{\mathbf{t}}/ds = \kappa \hat{\mathbf{n}}$, with $\kappa \geq 0$ the curvature and $\hat{\mathbf{n}}$ the unit principal normal, and since for any vector $\|\hat{\mathbf{t}} \times \mathbf{v}\| = \|\mathbf{v}_{\perp}\|$,

$$\kappa = \|\hat{\mathbf{t}} \times \nabla \ln \alpha\| = -\hat{\mathbf{n}} \cdot \nabla \ln \alpha, \quad \hat{\mathbf{n}} = -\frac{(\nabla \ln \alpha)_{\perp}}{\|(\nabla \ln \alpha)_{\perp}\|}. \quad (26)$$

The principal normal, which is the direction in which the ray is turning, is anti-parallel to the transverse component of $\nabla \ln \alpha$: rays bend towards regions of low wave speed, and do so more sharply the larger the fractional transverse velocity gradient. A ray travelling directly along $\nabla \alpha$ has zero curvature, which is why a vertical ray in Example 1 is straight.

In Example 1, $\nabla \ln \alpha = (g/\alpha)\hat{\mathbf{z}}$ and $\hat{\mathbf{t}} = (\sin \theta, \cos \theta)$, so eq. (26) gives

$$\kappa = \frac{g \sin \theta}{\alpha} = gp = \frac{1}{R}, \quad (27)$$

in agreement with the constant radius found there, and $\hat{\mathbf{n}} = (\cos \theta, -\sin \theta)$ points upwards, towards the slower material. The script `verify_examples5.py` also integrates eq. (19) through a model containing a localised low-velocity anomaly and confirms that the numerically measured curvature of the ray agrees with eq. (26) to better than 0.1%. The bending of rays towards low velocities is what produces the focusing and caustic seen in the lecture’s figure of ray tracing through a low-velocity region.

Example 3: The transport equation in one dimension. A thin elastic rod has density $\rho(x)$ and Young's modulus $E(x)$, both smooth, so that its longitudinal displacement satisfies

$$\rho \frac{\partial^2 u}{\partial t^2} - \frac{\partial}{\partial x} \left(E \frac{\partial u}{\partial x} \right) = 0.$$

(a) Insert the one-dimensional ray series $u(x, \omega) = \sum_{n=0}^{\infty} a^n(x) (i\omega)^{-n} e^{-i\omega T(x)}$ into the frequency-domain equation of motion and, following the lecture's derivation, obtain the eikonal equation and the equations satisfied by a^0 , a^1 and the general a^n .

(b) Solve the eikonal equation, and show that the first transport equation gives $a^0 \propto (\rho c)^{-1/2}$ with $c = \sqrt{E/\rho}$.

(c) Interpret this result in terms of energy flux, and explain what goes wrong if the impedance ρc has a jump discontinuity.

Solution. (a) In the frequency domain, with the convention $\partial_t \rightarrow i\omega$ used in the lecture, the equation of motion becomes

$$(i\omega)^2 \rho u - \frac{d}{dx} \left(E \frac{du}{dx} \right) = 0, \quad (28)$$

where u now denotes the Fourier transform. The one-dimensional slowness is $T' = dT/dx$, which plays the role of $p_i = \partial T / \partial x_i$ in the lecture. Differentiating the ray series term by term,

$$\frac{du}{dx} = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} (a^{n'} - i\omega T' a^n) e^{-i\omega T}, \quad (29)$$

$$\frac{d}{dx} \left(E \frac{du}{dx} \right) = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left[(E a^{n'})' - i\omega (E T' a^n)' - i\omega E T' a^{n''} + (i\omega)^2 E T'^2 a^n \right] e^{-i\omega T}, \quad (30)$$

where a prime on a^n or on a bracket denotes d/dx , and the two terms proportional to $i\omega$ in eq. (30) come respectively from differentiating the amplitude factor $E T' a^n$ and from differentiating the exponential in the term $E a^{n'} e^{-i\omega T}$. These are the one-dimensional counterparts of the lecture's expressions for $\partial u_k / \partial x_l$ and for the divergence of the stress. Substituting into eq. (28) and multiplying by -1 ,

$$\sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left[(E a^{n'})' - i\omega (E T' a^n)' - i\omega E T' a^{n''} + (i\omega)^2 (E T'^2 - \rho) a^n \right] e^{-i\omega T} = 0. \quad (31)$$

This must hold for all ω , so the coefficient of each power of $i\omega$ vanishes. The highest power, $(i\omega)^2$, arises only from the $n = 0$ term and gives

$$(E T'^2 - \rho) a^0 = 0, \quad (32)$$

the coefficient of $(i\omega)^1$ collects the last term with $n = 1$ and the middle terms with $n = 0$,

$$(E T'^2 - \rho) a^1 = (E T' a^0)' + E T' a^{0'}, \quad (33)$$

and for $(i\omega)^{2-n}$ with $n \geq 2$ three successive terms of the series contribute,

$$(ET'^2 - \rho) a^n = (ET' a^{n-1})' + ET' a^{n-1'} - (E a^{n-2'})'. \quad (34)$$

These are exactly the lecture's three equations with $A_{ijkl} \rightarrow E$, $p_i \rightarrow T'$ and $\delta_{ik} \rightarrow 1$. For a non-trivial a^0 , eq. (32) requires the eikonal equation

$$ET'^2 = \rho, \quad \text{i.e.} \quad T' = \pm \frac{1}{c}, \quad c = \sqrt{\frac{E}{\rho}}. \quad (35)$$

In one dimension the local Christoffel operator is the scalar $\Gamma = ET'^2/\rho = c^2 T'^2$, the ray Hamiltonian is $H = \frac{1}{2} c^2 p^2$, and the local slowness surface consists of the two points $p = \pm 1/c$; there is no polarisation to determine. The important structural difference from three dimensions now appears: once eq. (35) holds, the left hand sides of eqs. (33) and (34) vanish identically, and so every one of these equations is a constraint on the previous amplitude. Thus

$$2ET' a^{0'} + (ET')' a^0 = 0, \quad (36)$$

which is the first transport equation, and for $n \geq 1$

$$2ET' a^{n'} + (ET')' a^n = (E a^{n-1'})', \quad (37)$$

so each higher amplitude satisfies the same first-order equation, forced by the second derivative of its predecessor.

(b) Taking the right-going wave, $T' = 1/c$, the travel time is $T(x) = \int_0^x dx'/c(x')$ and $ET' = E/c = \sqrt{\rho E} = \rho c$. Eq. (36) then reads

$$\frac{a^{0'}}{a^0} = -\frac{1}{2} \frac{(\rho c)'}{\rho c}, \quad \text{so that} \quad a^0(x) = \frac{C}{\sqrt{\rho(x)c(x)}}, \quad (38)$$

with C a constant. Equivalently, multiplying eq. (36) by a^0 gives the conservation law

$$\frac{d}{dx} [ET' (a^0)^2] = \frac{d}{dx} [\rho c (a^0)^2] = 0. \quad (39)$$

The zeroth-order ray solution is therefore

$$u(x, \omega) \approx \frac{C}{\sqrt{\rho(x)c(x)}} \exp \left[-i\omega \int_0^x \frac{dx'}{c(x')} \right], \quad (40)$$

which you may recognise as the WKB approximation.

(c) For the time-harmonic wave $u = a \cos \omega(t - T)$ the energy flux along the rod is $-E \partial_x u \partial_t u$. To leading order in ω we have $\partial_x u \approx (a\omega/c) \sin \omega(t - T)$ and $\partial_t u = -a\omega \sin \omega(t - T)$, and so the flux averaged over a period is

$$\langle F \rangle = \frac{1}{2} \frac{E}{c} \omega^2 a^2 = \frac{1}{2} \rho c \omega^2 a^2. \quad (41)$$

Eq. (39) is therefore the statement that the energy flux carried by the wave is independent of position: in a smoothly varying rod nothing is reflected at leading order,

and the amplitude adjusts so that the same power is transmitted through every cross-section. The quantity $Z = \rho c = \sqrt{\rho E}$ is the acoustic impedance, and it is only through Z that the material affects the amplitude; a rod in which c varies but ρc is constant has a^0 constant.

If Z jumps from Z_1 to Z_2 at some point, eq. (38) would predict that the transmitted amplitude is $\sqrt{Z_1/Z_2}$ times the incident amplitude, with a^0 discontinuous and $a^{0'}$ containing a delta function, so that the assumption of smooth amplitude functions underlying the ray series fails. The exact matching conditions (continuity of u and of $E \partial_x u$) instead give a transmitted amplitude $2Z_1/(Z_1 + Z_2)$ together with a reflected wave of relative amplitude $(Z_1 - Z_2)/(Z_1 + Z_2)$ whose phase $T' = -1/c$ belongs to the other point of the local slowness surface. A single ray series carries only one phase function and so cannot represent this reflected wave; a discontinuity has to be handled by matching separate ray series across it, as remarked at the end of the lecture. Writing $Z_2 = Z_1(1 + \delta)$, both $\sqrt{Z_1/Z_2}$ and $2Z_1/(Z_1 + Z_2)$ equal $1 - \delta/2 + O(\delta^2)$, so the ray-theoretic amplitude is correct precisely to the order at which reflection can be neglected. As a numerical check, `verify_examples5.py` integrates eq. (28) exactly through a smooth rod and compares with eq. (40): the relative error is 0.8%, 0.4% and 0.2% at $\omega = 20, 40$ and 80 (in units where the rod has length 10 and $c \approx 3$), halving each time the frequency doubles, as expected for a neglected term of order $1/\omega$.

Example 4: Method of characteristics for a point source. A homogeneous and isotropic medium in two dimensions has constant P-wave speed α and density ρ . A source at the origin is modelled by the initial data $T = 0$ at $\mathbf{x} = \mathbf{0}$, so that the initial surface S^0 in phase space consists of the points $(\mathbf{x}^0, \mathbf{p}^0) = (\mathbf{0}, \hat{\mathbf{p}}/\alpha)$ with $\hat{\mathbf{p}} = (\cos \phi, \sin \phi)$ for all $\phi \in [0, 2\pi)$. (a) Integrate Hamilton's equations from this initial data to show that $T(\mathbf{x}) = \|\mathbf{x}\|/\alpha$, verify that this satisfies the eikonal equation, and check the symmetry $\partial p_i / \partial x_j = \partial p_j / \partial x_i$ of the resulting slowness field. (b) Show that for a homogeneous isotropic medium the $i\omega$ equation of the ray series, projected onto $\hat{\mathbf{p}}$, gives the transport equation $\nabla \cdot (A_0^2 \nabla T) = 0$ for the scalar P-wave amplitude A_0 defined by $\mathbf{a}^0 = A_0 \hat{\mathbf{p}}$. Solve it for the point source and comment on the behaviour at the origin.

Solution. (a) With α constant, eq. (19) reduces to $d\mathbf{x}/d\sigma = \alpha^2 \mathbf{p}$ and $d\mathbf{p}/d\sigma = \mathbf{0}$, so from the initial point labelled by ϕ we obtain

$$\mathbf{p}(\sigma) = \frac{\hat{\mathbf{p}}}{\alpha}, \quad \mathbf{x}(\sigma) = \alpha\sigma \hat{\mathbf{p}} = \alpha\sigma(\cos \phi, \sin \phi). \quad (42)$$

The rays are straight lines radiating from the origin, in accordance with Example 2. Given any $\mathbf{x} \neq \mathbf{0}$, there is exactly one ray through it, namely that with ϕ equal to the polar angle of $\mathbf{x}$, and the point is reached when $\sigma = \|\mathbf{x}\|/\alpha$. Since the generating parameter is the travel time,

$$T(\mathbf{x}) = \frac{\|\mathbf{x}\|}{\alpha} = \frac{r}{\alpha}, \quad (43)$$

whose wavefronts $T = \text{constant}$ are circles centred on the source. Directly, $\nabla T = \hat{\mathbf{x}}/\alpha$ and so $\|\nabla T\| = 1/\alpha$, which is the isotropic eikonal equation of Example 1. The

slowness field constructed by the method of characteristics is $p_i(\mathbf{x}) = x_i/(\alpha r)$, with

$$\frac{\partial p_i}{\partial x_j} = \frac{1}{\alpha r} (\delta_{ij} - \hat{x}_i \hat{x}_j), \quad (44)$$

which is symmetric in i and j as the lecture's general argument requires, so that $\mathbf{p}$ is indeed a gradient. Note that the initial surface S^0 here is a circle of slowness vectors sitting above a single point, so it is not the graph of a vector field over the initial data; the level surface $H = \frac{1}{2}$ swept out by the characteristics, $\{(\mathbf{x}, \hat{\mathbf{x}}/\alpha)\}$, becomes the graph of ∇T only away from the origin.

(b) For a P-wave $\hat{\mathbf{a}}^0 = \hat{\mathbf{p}}$, so we write

$$\mathbf{a}^0 = A_0 \hat{\mathbf{p}} = \varphi \mathbf{p}, \quad \varphi = \alpha A_0, \quad (45)$$

using $\hat{\mathbf{p}} = \alpha \mathbf{p}$ on the P-wave sheet. The lecture's equation at order $i\omega$ is

$$[A_{ijkl} p_j p_l - \rho \delta_{ik}] a_k^1 = \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^0) + A_{ijkl} p_j \frac{\partial a_k^0}{\partial x_l} \equiv F_i, \quad (46)$$

and since the matrix on the left is $\rho(\Gamma - \mathbf{I})$, which is symmetric and annihilates $\hat{\mathbf{p}}$, contracting with $\hat{p}_i$ removes the unknown $\mathbf{a}^1$ entirely and leaves the solvability condition $\hat{p}_i F_i = 0$. To evaluate it we use two facts about the slowness field: $\partial_j p_i = \partial_i p_j$, and, because $\|\mathbf{p}\| = 1/\alpha$ is constant in a homogeneous medium, $p_i \partial_j p_i = \frac{1}{2} \partial_j \|\mathbf{p}\|^2 = 0$, which together imply $p_j \partial_j p_i = p_j \partial_i p_j = 0$ as well. With $A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$ constant, the first term of F_i contracted with $\hat{p}_i$ is

$$\hat{p}_i A_{ijkl} \partial_j (\varphi p_l p_k) = \lambda \hat{p}_j \partial_j (\varphi \|\mathbf{p}\|^2) + 2\mu \hat{p}_k \partial_l (\varphi p_l p_k) = \frac{\lambda}{\alpha^2} \hat{\mathbf{p}} \cdot \nabla \varphi + \frac{2\mu}{\alpha} \nabla \cdot (\varphi \mathbf{p}), \quad (47)$$

where the two μ terms are equal by the symmetry of $p_l p_k$, and where we have expanded $\hat{p}_k \partial_l (\varphi p_l p_k) = \hat{p}_k p_k \partial_l (\varphi p_l) + \varphi p_l \hat{p}_k \partial_l p_k$ and used $\hat{p}_k p_k = 1/\alpha$ and $p_l \partial_l p_k = 0$. The second term gives

$$\begin{aligned} \hat{p}_i A_{ijkl} p_j \partial_l (\varphi p_k) &= \lambda \hat{p}_j p_j \partial_k (\varphi p_k) + \mu \hat{p}_k p_j \partial_j (\varphi p_k) + \mu \hat{p}_l p_j \partial_l (\varphi p_j) \\ &= \frac{\lambda}{\alpha} \nabla \cdot (\varphi \mathbf{p}) + \frac{2\mu}{\alpha^2} \hat{\mathbf{p}} \cdot \nabla \varphi, \end{aligned} \quad (48)$$

since $\hat{p}_k p_j \partial_j (\varphi p_k) = \hat{p}_k p_k p_j \partial_j \varphi = \alpha^{-2} \hat{\mathbf{p}} \cdot \nabla \varphi$ and $\hat{p}_l p_j \partial_l (\varphi p_j) = \|\mathbf{p}\|^2 \hat{p}_l \partial_l \varphi = \alpha^{-2} \hat{\mathbf{p}} \cdot \nabla \varphi$. Adding eqs. (47) and (48), and using $\hat{\mathbf{p}} = \alpha \mathbf{p}$,

$$\hat{p}_i F_i = \frac{\lambda + 2\mu}{\alpha} [\mathbf{p} \cdot \nabla \varphi + \nabla \cdot (\varphi \mathbf{p})] = \frac{\lambda + 2\mu}{\alpha} [2\mathbf{p} \cdot \nabla \varphi + \varphi \nabla \cdot \mathbf{p}] = \frac{\lambda + 2\mu}{\alpha \varphi} \nabla \cdot (\varphi^2 \mathbf{p}). \quad (49)$$

Setting this to zero, and recalling that $\mathbf{p} = \nabla T$ and that $\varphi = \alpha A_0$ with α constant, we arrive at the transport equation

$$\nabla \cdot (A_0^2 \nabla T) = 0. \quad (50)$$

This is the two- and three-dimensional counterpart of eq. (39). In a heterogeneous isotropic medium the same calculation, with rather more bookkeeping, gives the transport equation in the form

$$\nabla \cdot (\rho \alpha^2 A_0^2 \nabla T) = 0, \quad (51)$$

of which eq. (39) is the one-dimensional case with $\rho c^2 = E$.

For the point source $\nabla T = \hat{\mathbf{x}}/\alpha$, and in plane polar coordinates eq. (50) becomes

$$\frac{1}{r} \frac{d}{dr} (r A_0^2) = 0, \quad \text{so that} \quad A_0(r) = \frac{C}{\sqrt{r}}, \quad (52)$$

while in three dimensions the same argument with spherical wavefronts gives $A_0 \propto 1/r$. Physically, $\rho \alpha \omega^2 A_0^2$ times the length $2\pi r$ of the wavefront is the power crossing it, and this must be the same for every r . The amplitude is infinite at $r = 0$, where all of the rays intersect: the source point is a caustic, in fact the most degenerate kind, and ray theory can describe the field radiated by the source but not the source region itself. In the script `verify_examples5.py` the full right hand side F_i of eq. (46) is evaluated by finite differences for the isotropic tensor with $\mathbf{a}^0 = r^{-1/2} \hat{\mathbf{p}}$, and $\hat{p}_i F_i$ is found to vanish to rounding error, whereas it does not for $\mathbf{a}^0 = r^{-1} \hat{\mathbf{p}}$.

Example 5: Where is the approximation? For a plane wave entering a region in which the material parameters vary smoothly over a length scale L , estimate the size of the neglected $n = 1$ term of the ray series relative to the $n = 0$ term. Express the result in terms of the frequency ω and the time $\tau = L/\alpha$ taken by the wave to cross the heterogeneity, and in terms of the wavelength λ . Evaluate it for $L = 1000 \text{ km}$, $\alpha = 10 \text{ km s}^{-1}$ and a period of 20 s, and again for $L = 50 \text{ km}$.

Solution. The ratio of successive terms in the ray series is $|\mathbf{a}^1|/(\omega|\mathbf{a}^0|)$, so we need the size of $\mathbf{a}^1$, which is fixed by the order $i\omega$ equation, eq. (46). On the left, $A_{ijkl} p_j p_l$ and $\rho \delta_{ik}$ are each of order ρ , since $A_{ijkl} \sim \rho \alpha^2$ and $\|\mathbf{p}\| = 1/\alpha$. On the right, each term contains one spatial derivative acting on a product of A_{ijkl} , $\mathbf{p}$ and $\mathbf{a}^0$, all of which vary over the length scale L of the heterogeneity (the travel time and amplitude inherit their length scale from the medium), and so

$$\rho |\mathbf{a}^1| \sim \frac{\rho \alpha^2}{\alpha} \frac{|\mathbf{a}^0|}{L}, \quad \text{i.e.} \quad |\mathbf{a}^1| \sim \frac{\alpha}{L} |\mathbf{a}^0|. \quad (53)$$

The matrix on the left of eq. (46) is singular, so strictly this fixes only the components of $\mathbf{a}^1$ orthogonal to $\hat{\mathbf{a}}^0$; the remaining component is fixed by the solvability condition of the $n = 2$ equation, which has the same structure and gives the same estimate. The one-dimensional case makes this concrete: eq. (37) with $n = 1$ reads $2\rho c a^{1'} + (\rho c)' a^1 = (E a^{0'})'$, whose right hand side is of order $E a^0/L^2$, so that over a distance L we obtain $a^1 \sim E a^0/(\rho c L) = c a^0/L$. Hence the relative size of the first correction is

$$\frac{|\mathbf{a}^1|}{\omega |\mathbf{a}^0|} \sim \frac{\alpha}{\omega L} = \frac{1}{\omega \tau} = \frac{\lambda}{2\pi L}, \quad (54)$$

where we have used $\omega = 2\pi\alpha/\lambda$. The ray series is thus an expansion in the number of wavelengths, over 2π , that fit inside the heterogeneity, or equivalently in the number of radians of phase the wave accumulates while crossing it; this is the quantitative form of the condition $L/\lambda \gg 1$ under which the lecture introduced ray theory. For the first set of numbers, $\lambda = 200 \text{ km}$, $\tau = 100 \text{ s}$, $\omega = 0.314 \text{ s}^{-1}$ and $\omega\tau = 31.4$, so the neglected term is about 3% of the leading one and ray theory should work well; heterogeneity on this scale is typical of the lower mantle. For $L = 50 \text{ km}$, which is

more representative of crustal structure, the ratio is 0.64 and the expansion is useless at this period: the wave is scattered rather than refracted, and one is in the regime $L/\lambda \approx 1$ where the Born approximation, not ray theory, is the appropriate tool. Note finally that this estimate is local. Even when it is small, the errors accumulate along a ray, and it says nothing about the smooth, possibly large, error term discussed in the lecture, nor about caustics, where $\mathbf{a}^0$ itself fails.